

Electricity and Magnetism — Workout

ICSE Class 7 Physics · 10 MCQs · 1 mark each · Name: _____ Date: _____

Syllabus: Whole chapter: magnetism and the sure test, electromagnets and their uses, sources of electricity, cells in series, circuit symbols, and series & parallel circuits.

Instructions: Each question has one correct option. Tick or circle the letter of the best answer. Attempt all questions.

1. The sure test that a given bar is a magnet is that it shows ____ with the like pole of another magnet.

- | | |
|--------------|---------------|
| A. heating | B. repulsion |
| C. no effect | D. attraction |

2. Reversing the current in an electromagnet reverses its:

- | | |
|--------------------|--------------------|
| A. poles (N and S) | B. number of turns |
| C. strength | D. core |

3. An electromagnet is preferred in a crane because, unlike a permanent magnet, it can:

- | | |
|---|-------------------------|
| A. be switched on to grip and off to release the load | B. lift wood |
| C. work without a core | D. never lose magnetism |

4. A cell is a source of electricity that changes ____ energy into electrical energy.

- | | |
|-------------|-------------|
| A. light | B. sound |
| C. magnetic | D. chemical |

5. Three 2 V cells joined in series give a total of:

- | | |
|--------|-----------|
| A. 8 V | B. 6 V |
| C. 2 V | D. 0.67 V |

6. The thin wire inside a bulb that glows is the:

- | | |
|-------------|-------------|
| A. filament | B. terminal |
| C. switch | D. fuse |

7. In a circuit diagram, a cell is shown by:

- | | |
|---------------|-------------------------------------|
| A. a triangle | B. one long line and one short line |
| C. a zig-zag | D. a circle with a cross |

8. Components on separate branches, each getting the full voltage, are connected in:

- | | |
|--------------------|-------------|
| A. a short circuit | B. series |
| C. a single loop | D. parallel |

9. In a series circuit of three bulbs, removing one bulb makes the others:

- | | |
|------------------|------------------|
| A. glow brighter | B. stay the same |
| C. go out | D. reverse |

10. A safe precaution before switching on a circuit is to keep the switch ____ while making the connections.

- | | |
|---------------|--------|
| A. closed | B. wet |
| C. open (off) | D. hot |

Answer Key

1. B

2. A

3. A

4. D

5. B

6. A

7. B

8. D

9. C

10. C